

Error Propagation in ECB and CBC Modes

20. Write a C program for ECB mode, if there is an error in a block of the transmitted ciphertext, only the corresponding plaintext block is affected. However, in the CBC mode, this error propagates. For example, an error in the transmitted C1 obviously corrupts P1 and P2.

a. Are any blocks beyond P2 affected?

b. Suppose that there is a bit error in the source version of P1. Through how many

ciphertext blocks is this error propagated? What is the effect at the receiver?

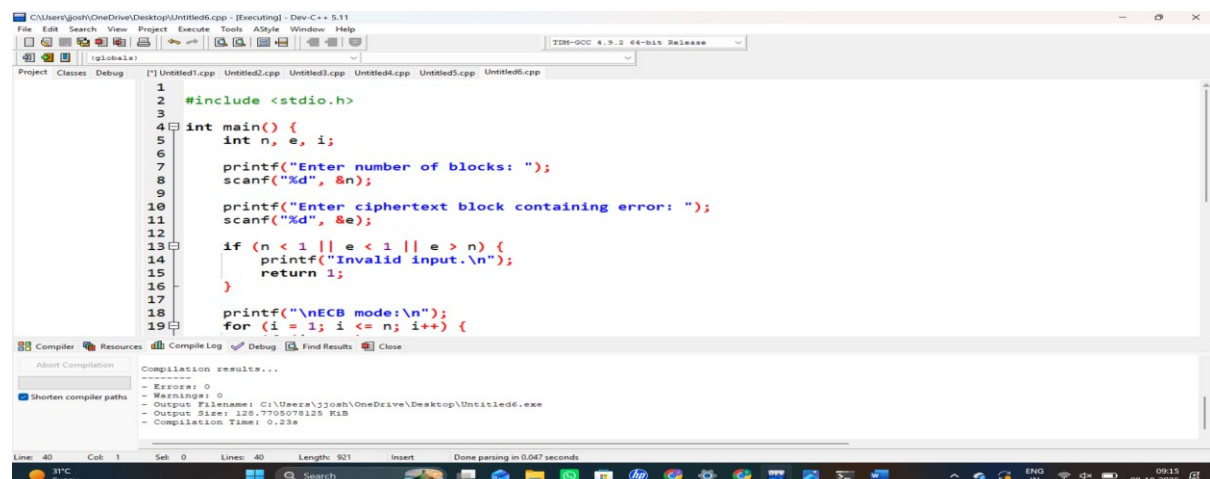
Aim:

To write a C program demonstrating the effect of a ciphertext block error in ECB and CBC modes and study error propagation.

Algorithm:

1. Start the program.
2. Read the total number of ciphertext blocks and the block containing an error.
3. In ECB mode, mark only the corresponding plaintext block as affected.
4. In CBC mode, mark the corresponding plaintext block and the next plaintext block as affected.
5. Display the affected blocks and explain the effect of an error in the source plaintext.
6. Stop.

Program :



```
1 #include <stdio.h>
2
3 int main() {
4     int n, e, i;
5
6     printf("Enter number of blocks: ");
7     scanf("%d", &n);
8
9     printf("Enter ciphertext block containing error: ");
10    scanf("%d", &e);
11
12    if (n < 1 || e < 1 || e > n) {
13        printf("Invalid input.\n");
14        return 1;
15    }
16
17    printf("\nECB mode:\n");
18    for (i = 1; i <= n; i++) {
```

The screenshot shows a C program in Dev-C++ for ECB mode. The code prompts the user for the number of blocks (n) and the block containing an error (e). It then checks for invalid input. For ECB mode, it prints the error for each block from 1 to n. The program is compiled successfully, as shown in the output window.

Input/Output:

```
C:\Users\josh\OneDrive\Desktop >
Enter number of blocks: 4
Enter ciphertext block containing error: 1

ECB mode:
P1: Affected
P2: Not affected
P3: Not affected
P4: Not affected

CBC mode:
P1: Affected
P2: Affected
P3: Not affected
P4: Not affected

Source plaintext P1 error in CBC:
Ciphertext blocks C1 to C4 may be affected.
The receiver may get corrupted plaintext blocks.

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Process exited after 3.43 seconds with return value 0
Press any key to continue . . .
```

Answers to the questions:

(a) Are any blocks beyond P2 affected by an error in transmitted C1 in CBC mode?

No. Only P1 and P2 are affected. P1 is corrupted by the damaged C1, and P2 is affected by the XOR operation with C1. Blocks P3 onward remain unaffected.

(b) If there is a bit error in the source version of P1, through how many ciphertext blocks does the error propagate? What is the effect at the receiver?

- Ciphertext: The error may propagate through all subsequent ciphertext blocks, from C1 to Cn, because CBC encryption chains each block to the previous ciphertext block.
- Receiver: The corresponding decrypted plaintext blocks may be corrupted throughout the remaining message.

Result:

The program demonstrates that ECB confines a ciphertext error to one plaintext block, whereas CBC causes an error in a ciphertext block to affect two plaintext blocks. A source plaintext error in CBC can propagate through the remaining ciphertext and plaintext blocks.